

Summary

Software Engineer specializing in Gen AI systems, RAG pipelines, and agentic workflows. **6x Hackathon Winner** including **Smart India Hackathon 2024**. Led a **\$4,000 IEEE HTB Grant Project** managing a team of 5 engineers. Strong in **Python**, with hands-on experience shipping **RAG systems**, **AI Agents**, and **Generative AI workflows**.

Technical Skills

- Programming Languages:** Python, C, C++, SQL
AI/ML & MLOps: Machine Learning, Generative AI, Agentic AI, LLMs, LangChain, RAG, Prompt Engineering, MCP, Vector Databases (FAISS, Qdrant), Transformers, Hugging Face, Mediapipe
Backend & APIs: FastAPI, REST APIs
Databases: PostgreSQL, Supabase
Developer Tools: Git, GitHub, Docker, VS Code, Android Studio
CS Fundamentals: DSA, DBMS, OOPs, Design Patterns, Debugging, SDLC, Software Testing, Operating Systems

Education

Acropolis Institute of Technology and Research

Indore, India

B.Tech. in Computer Science & Engineering (Artificial Intelligence & Machine Learning); CGPA: 7.43

July 2026

Experience

- SDE 1

May 2026 – Present

Vected Technologies Pvt. Ltd.

Indore, India

- **Worked on MLOps pipelines on Azure Databricks** to eliminate silent failures and improve deployment reliability.
 - **Developed RAG-based Gen AI applications** using **LangChain**, **vector databases**, and **LLMs**, enabling intelligent document retrieval and context-aware Q&A across enterprise knowledge bases.
 - **Designed AI Agents** with tool-calling and multi-step reasoning to automate internal business workflows, pulling data from external APIs and reducing repetitive manual effort.
- Tech Lead

Jul 2024 – Mar 2025

IEEE SIGHT

Indore, India

- **Led end-to-end development** of “Harvesting Hope,” a humanitarian AI+IoT platform awarded a **\$4,000 grant** by IEEE HTB to help farmers improve crop yields through AI-driven soil analysis and IoT sensors.
 - **Coordinated a team of 5 engineers** with **Git branching strategies** and structured **code reviews** to build and ship the *Krishi* mobile application.
 - **Trained a YOLOv8 model** on a custom dataset of 1,000+ images for real-time pest detection, achieving **90% accuracy** and integrating it into the production mobile app.

Projects

- Signify – Real-Time ISL Translation

- **Built award-winning (National Winner, Smart India Hackathon 2024)** real-time Indian Sign Language translation system achieving **90% accuracy** across multiple regional languages. Developed a **Flutter mobile app** for live video capture and translation display.
 - **Designed and coded ML pipeline** with optimized **MediaPipe** hand-tracking and **Random Forest classifiers**; deployed via scalable **REST API** architecture with proper **OOP design patterns**.
 - **Optimized inference latency** to under **200ms** for real-time performance.
- Sentient – AI NPC Engine

- **Built an LLM + RAG dialogue pipeline** that ingests game manuals, instructions, and character profiles to generate personalized NPC conversations dynamically instead of hardcoded scripts.
 - **Architected full-stack application** with **React.js** frontend and high-performance **FastAPI** backend communicating via secure **RESTful APIs**.
 - **Implemented semantic search** using **FAISS vector database** and **embedding models** for faster query response times.
- Vrinda - Smart Agriculture IoT App

- **Built an IoT-integrated mobile app** that connects to **ESP32 microcontroller** via Bluetooth to receive real-time **NPK and soil moisture** sensor readings, enabling data-driven crop management for farmers.
 - **Integrated LLM-based advisory engine** that processes live sensor data to generate personalized farming suggestions; added **weather forecasting**, agriculture news, and government scheme modules for a comprehensive farmer dashboard.
 - Deployed a **YOLOv8 real-time object detection model** to identify crop pests from camera input, delivering tailored management recommendations that improved crop yield by **20%** across 10+ users.

Honors & Awards

- **IEEE Tech4Good Grant Funded Project Tech Lead**
- **Hackathon Achievements (6 Wins):** National Winner – Smart India Hackathon (SIH) 2024, 1st Place – Intellify 3.0, 1st Place – HackWave, 2nd Place – Code for Bharat 2, 2nd Place – Codespire, 4th Place – Prayatna.